Worksheet 11 - The Chain Rule

MATH 1271

10/15/2026

1. For each function below, identify how to express it as a composite of elementary functions and then find its derivative:

   (a) $m(v) = \sin(v^2)\cos(v^3)$

   (b) $h(y) = \dfrac{\cos(10y)}{e^{4y}+1}$

   (c) $g(w) = (6w^2+7w)^4$

   (d) $\alpha(x) = \left(\sqrt[3]{12e^{4x}+3e^{4x}+2} + \sin^2(e^{4x})\right)^{-1}$

2. Use known derivative rules, including the Chain Rule, as needed to answer each of the following questions:

   (a) Find an equation for the tangent line to the curve $y = \sqrt{e^x + 3}$ at the point where $x = 0$.

   (b) If $s(t) = \frac{1}{(t^2+1)^3}$ represents the position function of a particle moving horizontally along an axis at time t, find the particle's instantaneous velocity at $t = 1$. Is the particle moving to the left or to the right at that instance? (Hint: What is the sign of $v(1)$?)

   (c) At sea level, the air pressure is 30 inches of mercury. At an altitude of h feet above sea level, the air pressure, P, is given by the function $P = 30e^{-0.0000323h}$. Compute $\frac{dP}{dh}$ and explain what this derivative function tells you about air pressure, including a discussion of units of $\frac{dP}{dh}$.

(d) Referring to the previous question, determine how fast the air pressure is changing for a pilot of a small plane passing through an altitude of $1,000$ feet. (Round your answer to 6 decimal places.)

(e) Suppose that $f(x)$ and $g(x)$ are differentiable functions and that the following information about them is known:

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	2	-5	-3	4
2	-3	4	-1	2

If $C(x)$ is a function given by the formula $C(x) = f(g(x))$, determine $C'(2)$.

(f) Using the same table as in the previous question, let $D(x)$ be the function $f(f(x))$. Find $D'(-1)$.

3. Say you are working for a company who produces solar panels, and are trying to determine the optimal density of photovoltaic cells with which to produce your panels. Your economists have determined that clients are willing to pay more for panels as the energy density (in watts per square meter) of your panels vary; your chemists have determined that the energy density changes depending on the density of your photovoltaic cells, and that the cost per unit area product (dollars per square meter) also only depends on that photovoltaic density (measured in cells per square meter).

   Suppose you know that:

   - Your company currently produces panels with a PVD of 100 cells per square meter
   - At a PVD of 100 cells per square meter, the energy density produced 175 watts per meter squared
   - At that same PVD, panels cost about 60 dollars per square meter
   - At an energy density of 175 watts per meter squared, clients are willing to pay 90 dollars per square meter
   - At your current PVD, small changes to the PVD cause the energy density to change by about 0.02 watts per cell
   - At your current PVD, small changes to it will increase the cost by about 0.20 USD per cell
   - From market research, increasing energy density will make clients increase demand at a rate of about 2 dollars per watt.

   (a) If the energy density as a function of photovoltaic density is given by $E(p)$, the cost as a function of photovoltaic density is given by $C(p)$, and the price a client is willing to pay as a function of energy density is given by $W(e)$, determine an expression for the profit per square meter after selling a solar panel.

   (b) At our current values, how much profit are we making per square meter?

(c) Using the previously defined functions, rewrite the measurements on the previous page in the form of values of function/their derivatives.

(d) What is an expression for the derivative of profit as a function of photovoltaic density?

(e) Should you increase or decrease photovoltaic density if you want to increase profit?